

SQL, Database & Data Quality

Data validation, SQL, relational/NoSQL databases, ETL/ELT and pipeline quality.

Databases for QA

QA uses databases to validate persistence, integrations, business rules and consistency between UI, API and storage. Query safely and avoid unintended data changes.

Essential SQL

SELECT reads data; WHERE filters; ORDER BY sorts; GROUP BY aggregates; JOIN combines tables. Before UPDATE or DELETE, confirm environment, permissions and impact.

JOIN in practice

INNER JOIN returns matching rows. LEFT JOIN keeps all rows from the left and adds matches from the right. They are useful for validating orders, customers and relationships.

Transactions and ACID

Atomicity, Consistency, Isolation and Durability help explain guarantees in transactional databases. Concurrent scenarios and rollback deserve attention in critical operations.

Data Quality

Data quality includes dimensions such as completeness, accuracy, consistency, uniqueness and freshness. A system may work technically while still delivering wrong data.

ETL, ELT, Data Lake and Warehouse

Pipelines move and transform data between sources and destinations. QA can validate counts, schema, transformations, duplicates, reconciliation and freshness.

NoSQL

NoSQL databases use models different from relational databases, such as documents and key-value. Testing should consider consistency, flexible schema and query patterns.

Test-data security

Avoid real personal data when it is not necessary. Prefer synthetic data, masking and anonymization according to applicable policies and laws.

QA tip

Compare data at different points in the flow; a correct UI value does not prove the entire integration is correct.